COMS3003A
Test 1

Zachary Bowditch

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Total: 12

1. Give a regular expression describing the language of all strings from $\{a, b\}$ containing $abba$ as a substring (a string is also a substring of itself). **(1 mark)**

2. Let $\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Describe, in words, the kind of numbers generated by the following regular expression: $1\Sigma^2\{0, 5\}$ **(2 marks)**

3. For each of the following languages, if the language is regular, draw the state diagram of a finite-state machine that recognises it. Otherwise, prove its non-regularity using the Pumping Lemma **(9 marks)**:

   (a) All strings from $\{a, b\}$ **not** containing aba as a substring.

   (b) $\{a^{pq} \mid p, q \text{ are primes}\}$.

   (c) $\{\omega \mid \omega \text{ is a string in } (abb^*abb^*)^*\}$.

Lemma 1 (Pumping Lemma). *If R is a regular language, then there exists an integer $p \geq 1$ such that for every string $\omega \in R$ with $|w| \geq p$, ω can be written as $\omega = xyz$ where $|y| > 0$, $|xy| \leq p$ and $xy^iz \in R$, for each $i \geq 0$.*